

EtherChannel

- Topics covered:
 - What is EtherChannel? What problem does it solve?
 - Configuring Layer 2 (switch) EtherChannel
 - Configuring Layer 3 (switch) EtherChannel

EtherChannel

What EtherChannel Is and What Problem It Solves

EtherChannel is used to solve a major problem caused by Spanning Tree Protocol.

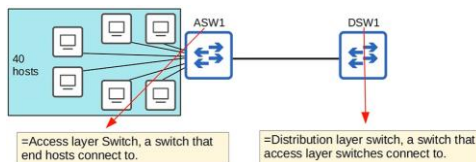
We will also look at:

- Layer 2 EtherChannel
- Layer 3 EtherChannel

A Layer 2 EtherChannel is a group of switch ports which operate as a single interface.
A Layer 3 EtherChannel is a group of routed ports which operate as a single interface, and you assign an IP address to it because it is Layer 3.

Network Used for Demonstration

We have two switches:



- **ASW1** → Access Switch (end hosts connect here)
- **DSW1** → Distribution Switch (access switches connect here)

There are many end hosts (40 hosts) connected to ASW1, and they are all trying to access the Internet.

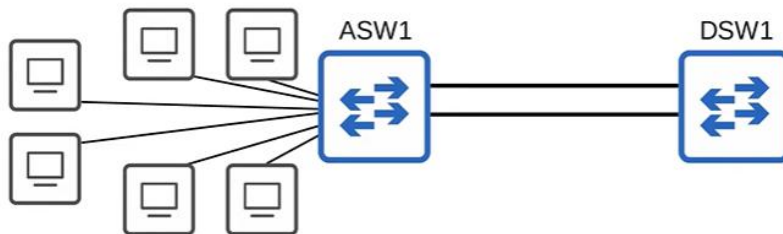
Explanation:

ASW1 connects end hosts. DSW1 connects access layer switches.

Congestion Problem

The connection between ASW1 and DSW1 becomes congested, so the network administrator decides to add another link to increase bandwidth.

After Adding One More Link:



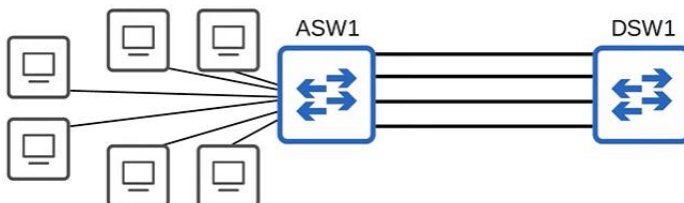
Explanation:

Another physical link is added to increase bandwidth.

Still Congested → More Links Added

The admin adds:

- a third link
- then a fourth link



Explanation:

Even with four physical links, congestion does not improve.

Oversubscription

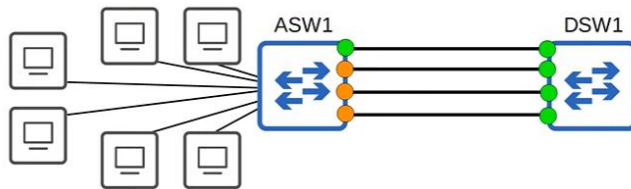
The total bandwidth of end host interfaces is greater than the bandwidth of the uplink, which is called oversubscription.

Some oversubscription is acceptable, but too much causes congestion.

Why Bandwidth Did Not Increase

The admin checks port lights.

- On **DSW1** → all lights are green
- On **ASW1** → only one green, others orange



Explanation:

Spanning Tree Protocol disables all but one link can transfer.

Role of Spanning Tree Protocol

If two switches are connected using multiple links:

- Only one link forwards
- Other links are blocked (orange)

This prevents:

- Layer 2 loops
- Broadcast storms

Other links are unused unless the active link fails.

Problem With STP Blocking:

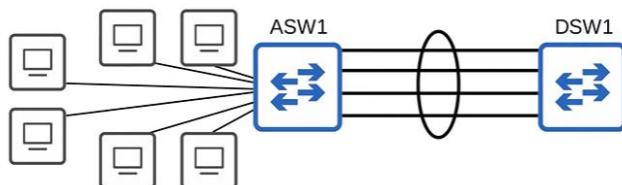
- Backup links are good
- But blocked links waste bandwidth
- Three interfaces are not forwarding any traffic

EtherChannel Solution

By forming **four physical interfaces into one logical interface**, EtherChannel provides:

- Redundancy
- Increased bandwidth

EtherChannel Representation:



Explanation:

The circle represents interfaces grouped into one EtherChannel.

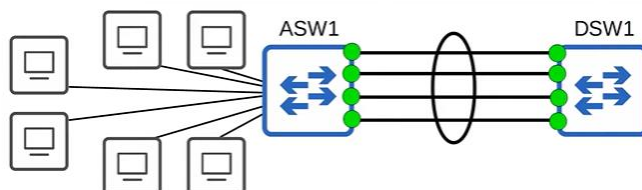
EtherChannel Behaviour

EtherChannel groups multiple interfaces to act as one single interface.

STP treats this group as one interface.

After forming EtherChannel:

- All port lights turn green



Explanation:

No ports are blocked because STP sees only one logical link.

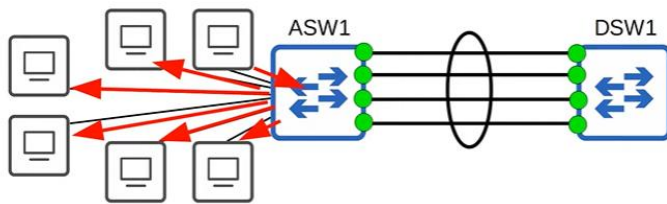
No Layer 2 Loop:

Even with four physical links:

- They behave as **one link**
- No Layer 2 loop is formed

Broadcast Frame Example

A PC sends a broadcast frame.



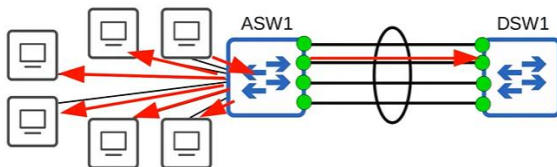
ASW1 floods the broadcast to all its PCs.

How Many Copies Does DSW1 Receive?

Although there are four physical interfaces:

- They behave as one interface
- DSW1 receives only one copy

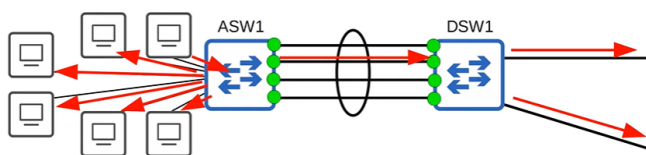
ASW1 does not send four copies of the same broadcast.



DSW1 Flooding Behaviour?

DSW1 floods the broadcast:

- Out of all interfaces except the one it was received on



Explanation: DSW1 does not forward the frame back into the EtherChannel.

Why Is No Loop Formed?

Although EtherChannel has four physical interfaces:

- They behave as one interface
- Frame is not sent back to ASW1
- No Layer 2 loop occurs

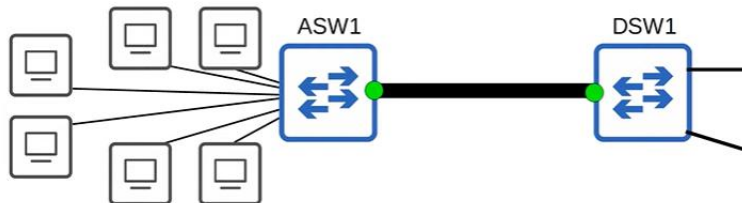
Logical vs Physical View

Instead of:

- Four separate gigabit interfaces

It behaves like:

- One four-gigabit virtual interface



Physical vs Virtual Concept

Just like VLANs:

- VLANs virtually divide PCs into separate LANs
- EtherChannel virtually combines interfaces into one

Other Names for EtherChannel

- Port Channel
- LAG (Link Aggregation Group)

Cisco IOS uses different terms during configuration.

EtherChannel Load Balancing

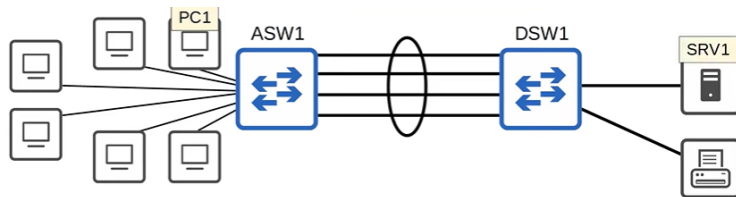
EtherChannel load balances traffic based on flows.

Flow Definition:

A flow is a communication between two nodes.

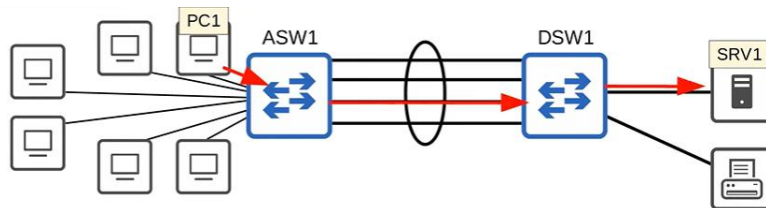
Example:

- PC1 ↔ SRV1



Load Balancing Example (PC1 → SRV1)

PC1 sends frames to SRV1.



PC1 → ASW1(+--- Physical Interface #3) → Port-Channel → DSW1 → SRV1

Explanation:

An algorithm selects one physical interface.

If PC1 send again frame to SRV1 then, all frames in the same flow use the same interface.

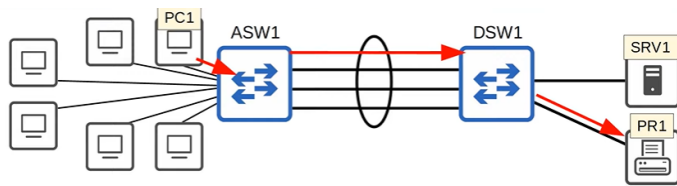
Why Is Same Interface Used?

If frames used different interfaces:

- Frames may arrive out of order
- Some applications cannot handle this

Second Flow (PC1 → PR1)

PC1 initiates another flow to PR1.



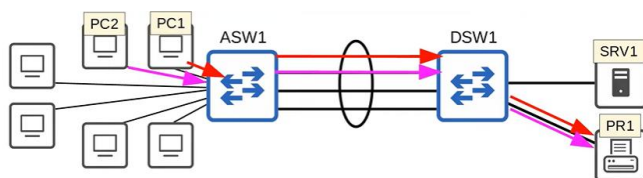
PC1 → ASW1(+--- Physical Interface #1) → Port-Channel → DSW1 → PR1

Explanation:

Different flow → separate calculation → different interface.

Third Flow (PC2 → PR1)

PC2 starts a print job.



PC2 → ASW1(+--- Physical Interface #2) → Port-Channel → DSW1 → PR1

Explanation:

Different source → different flow → different interface.

EtherChannel Load Balancing Summary

- Same flow → same physical interface
- Different flows → possibly different interfaces
- Load is distributed across interfaces

Load Balancing Inputs

The calculation can use:

- Source MAC address
- Destination MAC address
- Source + Destination MAC address
- Source IP address
- Destination IP address
- Source + Destination IP address

Some switches also support:

- Layer 4 TCP/UDP port numbers

Important Notes:

- Supported methods depend on switch model
- Some switches support only MAC-based methods
- Some support MAC and IP methods
- Some support all methods

Final Key Takeaway

EtherChannel:

- Combines multiple physical interfaces
- Acts as one logical interface
- Prevents STP blocking
- Increases bandwidth
- Provides redundancy
- Load balances traffic using flows

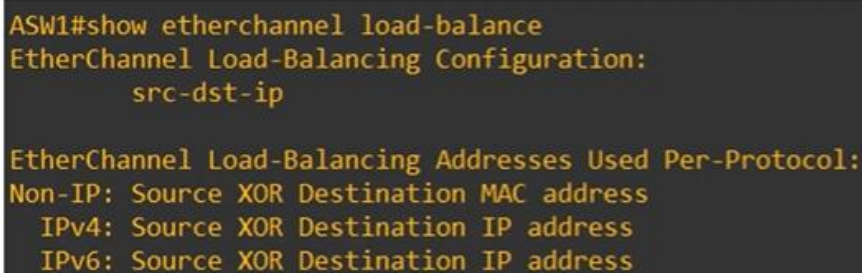
Configuring Layer 2 (switch) EtherChannel

Checking EtherChannel Load-Balancing Method

We haven't configured an EtherChannel yet, but before doing that we check how load-balancing works.

Command to Check Load-Balancing:

#show etherchannel load-balance



```
ASW1#show etherchannel load-balance
EtherChannel Load-Balancing Configuration:
      src-dst-ip

EtherChannel Load-Balancing Addresses Used Per-Protocol:
Non-IP: Source XOR Destination MAC address
IPv4:   Source XOR Destination IP address
IPv6:   Source XOR Destination IP address
```

This command shows the current load-balancing method used by the switch.

Default Load-Balancing Method:

On this model of switch, the default load-balancing method is based on:

- Source IP address
- Destination IP address

So, for example:

- Traffic from 10.0.0.1 → 10.0.0.2
- Will always use the same physical interface inside the EtherChannel

IP vs Non-IP Traffic Load-Balancing:

Breakdown

- Frames with IPv4 or IPv6 packets
 - Load-balanced using source and destination IP addresses
- Non-IP frames
 - Load-balanced using source and destination MAC addresses

Reason:

If an IP packet is not encapsulated in the Ethernet frame, there is no IP address, so MAC addresses are used instead.

Changing the Load-Balancing Method

Command (Global Configuration Mode):

#port-channel load-balance src-dst-mac

```
ASW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
ASW1(config)#port-channel load-balance src-dst-mac
ASW1(config)#do show etherchannel load-balance
EtherChannel Load-Balancing Configuration:
    src-dst-mac

EtherChannel Load-Balancing Addresses Used Per-Protocol:
Non-IP: Source XOR Destination MAC address
IPv4: Source XOR Destination MAC address
IPv6: Source XOR Destination MAC address

ASW1(config)#
```

This command changes the load-balancing method to use the source and destination MAC addresses.

Confirming the Change:

#do show etherchannel load-balance

This confirms that the load-balancing configuration has been successfully changed.

Available Load-Balancing Choices on This Device

Command:

#port-channel load-balance ?

```
ASW1(config)#port-channel load-balance ?
dst-ip      Dst IP Addr
dst-mac      Dst Mac Addr
src-dst-ip   Src XOR Dst IP Addr
src-dst-mac  Src XOR Dst Mac Addr
src-ip       Src IP Addr
src-mac      Src Mac Addr

ASW1(config)#port-channel load-bala
```

The switch can load-balance using:

- MAC addresses
- IP addresses

And in both cases, it can use:

- Source
- Destination
- Source and Destination

Important Cisco Keyword Confusion

- To configure load-balancing:

```
SW(config)# port-channel load-balance method
```

- To view load-balancing:

```
SW# show etherchannel load-balance
```

Different keywords are used for the same feature, which is frustrating.

EtherChannel Configuration Methods

There are three methods to configure EtherChannel.

Method 1: PAgP (Port Aggregation Protocol)

Method 2: LACP (Link Aggregation Control Protocol)

Method 3: Static EtherChannel

Method 1: PAgP (Port Aggregation Protocol)

- Cisco proprietary protocol
- Can be used only on Cisco switches
- Cannot be used with switches from other vendors

What PAgP Does

- Dynamically negotiates creation and maintenance of EtherChannel
- Works similar to DTP for trunks
- Sends frames to the neighbour switch to check if it wants to form an EtherChannel

PAgP Modes

- Desirable
 - Actively tries to form an EtherChannel
- Auto
 - Forms EtherChannel only if the other side is desirable

PAgP Summary

- Auto + Auto → No EtherChannel
- Auto + Desirable → EtherChannel forms
- Desirable + Desirable → EtherChannel forms

Method 2: LACP (Link Aggregation Control Protocol)

- IEEE standard protocol (802.3ad)
- Works on Cisco and non-Cisco switches
- Dynamically negotiates EtherChannel
- Preferred method for EtherChannel

The exam topics list only LACP, but you should know all methods.

LACP Modes

- Active
 - Actively tries to form EtherChannel
- Passive
 - Forms EtherChannel only if the other side is active

LACP Summary

- Passive + Passive → No EtherChannel
- Active + Passive → EtherChannel forms
- Active + Active → EtherChannel forms

Important LACP Note:

Even if both sides are configured as passive:

- The port-channel interface is created
- But it does not function unless one side is active

Method 3: Static EtherChannel

- No protocol used
- Interfaces are manually forced to form EtherChannel
- Mode used: ON

Static EtherChannel Behaviour:

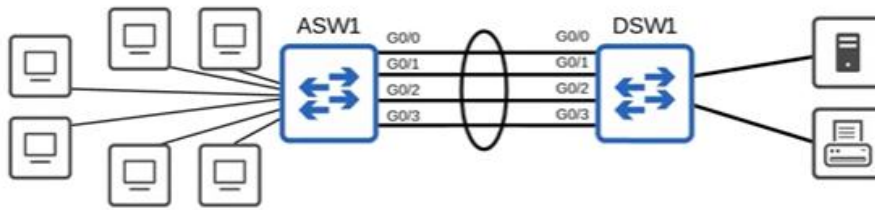
- ON + ON → EtherChannel forms
- ON + Desirable → No EtherChannel
- ON + Active → No EtherChannel

This method is usually avoided because the switch cannot dynamically maintain the EtherChannel.

Maximum Interfaces in EtherChannel

- Up to 8 interfaces can be active
- LACP allows 16 total
 - 8 active
 - 8 standby (used if an active link fails)

EtherChannel Configuration Process Of PAgP



```
ASW1(config)#interface range g0/0 - 3
ASW1(config-if-range)#channel-group 1 mode ?
    active      Enable LACP unconditionally
    auto        Enable PAgP only if a PAgP device is detected
    desirable   Enable PAgP unconditionally
    on          Enable Etherchannel only
    passive     Enable LACP only if a LACP device is detected

ASW1(config-if-range)#channel-group 1 mode desirable
Creating a port-channel interface Port-channel 1
```

Step 1: Interface Range Configuration

```
#interface range g0/0 - 3
```

Explanation: Configuring all member interfaces at once ensures all configurations match, which is required for EtherChannel.

Step 2: Channel-Group Command

```
#channel-group 1 mode desirable
```

Explanation:

- channel-group configures the EtherChannel
- The number identifies the virtual interface
- The mode determines the negotiation protocol

Step 3: Port-Channel Interface Creation

Verification Command: #SHOW IP INTERFACE BRIEF

```
GigabitEthernet5/2    unassigned    YES unset    up            up
GigabitEthernet2/2    unassigned    YES unset    up            up
Port-channel1         unassigned    YES unset    up            up
ASW1(config)#
```

Explanation: The virtual interface created is named port-channel, not channel-group.

Channel-Group Number Rule

- Must match between member interfaces on the same switch
- Does NOT need to match the number on the neighbouring switch

Example:

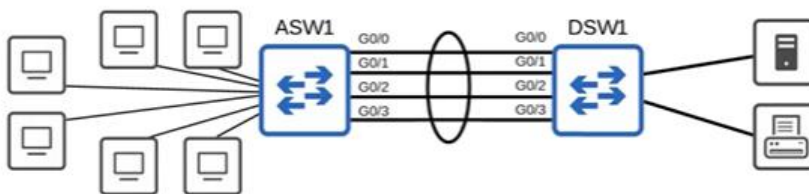
- ASW1 → channel-group 1
- DSW1 → channel-group 2

This still forms an EtherChannel.

LACP Configuration Difference

The command structure is the same, only mode names change.

#channel-group 1 mode active



```
ASW1(config-if-range)#channel-group 1 mode ?
active      Enable LACP unconditionally
auto        Enable PAgP only if a PAgP device is detected
desirable   Enable PAgP unconditionally
on          Enable Etherchannel only
passive      Enable LACP only if a LACP device is detected

ASW1(config-if-range)#channel-group 1 mode active
Creating a port-channel interface Port-channel 1
```

Explanation:

- active replaces desirable
- passive replaces auto

Static EtherChannel Configuration

Command: #channel-group 1 mode on

```
ASW1(config-if-range)#channel-group 1 mode ?
active      Enable LACP unconditionally
auto        Enable PAgP only if a PAgP device is detected
desirable   Enable PAgP unconditionally
on          Enable Etherchannel only
passive     Enable LACP only if a LACP device is detected

ASW1(config-if-range)#channel-group 1 mode on
Creating a port-channel interface Port-channel 1
```

Explanation: This manually forces interfaces to form an EtherChannel.

Final Key Points

- SHOW ETHERCHANNEL LOAD-BALANCE → check method
- PORT-CHANNEL LOAD-BALANCE → configure method
- CHANNEL-GROUP → creates EtherChannel
- Virtual interface name → port-channel
- PAgP → Cisco only
- LACP → IEEE standard, preferred
- Static ON mode → no negotiation

CHANNEL-PROTOCOL Command

Another command you should know is the CHANNEL-PROTOCOL command.

CLI Image:

```
ASW1(config-if-range)#channel-protocol ?  
lacp  Prepare interface for LACP protocol  
pagp  Prepare interface for PAgP protocol
```

Explanation: This command manually configures which negotiation protocol the member interfaces should use.

Why CHANNEL-PROTOCOL Is Not Very Useful?

This command isn't very useful because:

- channel-group 1 mode desirable / auto → automatically uses **PAgP**
- channel-group 1 mode active / passive → automatically uses **LACP**

So there isn't really a point in using this command, but it is important to know for the exam.

Protocol Mismatch Example

CLI Image:

```
ASW1(config-if-range)#channel-protocol lacp  
ASW1(config-if-range)#channel-group 1 mode desirable  
Command rejected (Channel protocol mismatch for interface Gi0/0 in group 1): the interface can not be added to the channel group  
  
% Range command terminated because it failed on GigabitEthernet0/0
```

Command rejected (Channel protocol mismatch)

Explanation: LACP was manually configured, but desirable mode is PAgP, so the command was rejected.

ON Mode with Protocol Mismatch

CLI Image:

```
ASW1(config-if-range)#channel-group 1 mode on  
Command rejected (Channel protocol mismatch for interface Gi0/0 in group 1): the interface can not be added to the channel group  
  
% Range command terminated because it failed on GigabitEthernet0/0
```

Command rejected (Channel protocol mismatch)

Explanation: ON mode was also rejected because the protocol did not match.

Correct LACP Mode

CLI Image:

```
ASW1(config-if-range)#channel-protocol ?
  lacp  Prepare interface for LACP protocol
  pagp  Prepare interface for PAgP protocol

ASW1(config-if-range)#channel-protocol lacp
ASW1(config-if-range)#channel-group 1 mode desirable
Command rejected (Channel protocol mismatch for interface Gi0/0 in group 1): the interface can not be added to the channel group

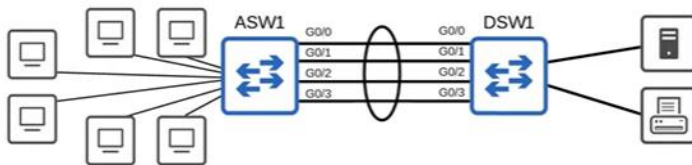
% Range command terminated because it failed on GigabitEthernet0/0
ASW1(config-if-range)#channel-group 1 mode on
Command rejected (Channel protocol mismatch for interface Gi0/0 in group 1): the interface can not be added to the channel group

% Range command terminated because it failed on GigabitEthernet0/0
ASW1(config-if-range)#channel-group 1 mode active
Creating a port-channel interface Port-channel 1
```

Creating a port-channel interface Port-channel1

Explanation: Active mode works because active is an LACP mode.

Configuring the Port-Channel Interface



After configuring the EtherChannel (PAgP, LACP, or static), you configure the port-channel interface itself.

Port-Channel Configured as Trunk:

CLI Image (Running-Config Section)

```
#interface Port-channel1
```

```
#switchport trunk encapsulation dot1q
```

```
#switchport mode trunk
```

```

ASW1(config)#interface port-channel 1
ASW1(config-if)#switchport trunk encapsulation dot1q
ASW1(config-if)#switchport mode trunk
ASW1(config-if)#do show interfaces trunk

```

Port	Mode	Encapsulation	Status	Native vlan
Po1	on	802.1q	trunking	1


```

Port      Vlans allowed on trunk
Po1       1-4094

Port      Vlans allowed and active in management domain
Po1       1

Port      Vlans in spanning tree forwarding state and not pruned
Po1       none

```

Explanation: Only The port-channel interface is configured as a trunk.

SHOW INTERFACES TRUNK Output:

Explanation: Only the port-channel interface appears. Individual physical interfaces are not listed.

Physical Interfaces Automatically Updated:

Here is the out section of #show running-config, The something interesting to notice here. The trunk configuration that I configured on the port-channel interface were also applied to the physical interfaces

CLI Image:

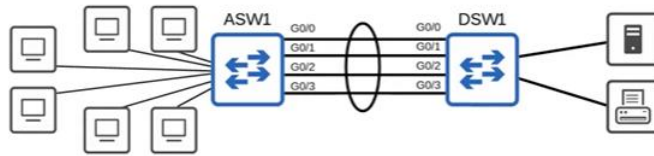
```

interface Port-channel1
  switchport trunk encapsulation dot1q
  switchport mode trunk
!
interface GigabitEthernet0/0
  switchport trunk encapsulation dot1q
  switchport mode trunk
  media-type rj45
  negotiation auto
  channel-protocol lacp
  channel-group 1 mode active
!
interface GigabitEthernet0/1
  switchport trunk encapsulation dot1q
  switchport mode trunk
  media-type rj45
  negotiation auto
  channel-protocol lacp
  channel-group 1 mode active
!
interface GigabitEthernet0/2
  switchport trunk encapsulation dot1q
  switchport mode trunk
  media-type rj45
  negotiation auto
  channel-protocol lacp
  channel-group 1 mode active
!
interface GigabitEthernet0/3
  switchport trunk encapsulation dot1q
  switchport mode trunk
  media-type rj45
  negotiation auto
  channel-protocol lacp
  channel-group 1 mode active
!

```

Explanation: Trunk configuration applied to the port-channel interface was also applied automatically to the physical interfaces.

Requirement for Member Interfaces



Member interfaces must have matching configurations:

- Same duplex
- Same speed
- Same switchport mode (access or trunk)
- Same allowed VLANs (if trunk)
- Same native VLAN (if trunk)

If configurations do not match, the interface is excluded from the EtherChannel.

Verifying EtherChannel Status

The most useful command is:

#SHOW ETHERCHANNEL SUMMARY

```
ASW1#show etherchannel summary
Flags: D - down          P - bundled in port-channel
       I - stand-alone  S - suspended
       H - Hot-standby (LACP only)
       R - Layer3       S - Layer2
       U - in use       N - not in use, no aggregation
       f - failed to allocate aggregator

       M - not in use, minimum links not met
       m - not in use, port not aggregated due to minimum links not met
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

       A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----
1      Po1(SU)       LACP       Gi0/24(P) Gi0/25(P) Gi0/26(P)
              Gi0/27(P)
```

Explanation:

- **S** = Layer 2 EtherChannel (switchport)
- **U** = In use
- **P** = Port properly bundled

These are the flags you want to see in an operational Layer 2 EtherChannel.

Port-Channel Shutdown Example:

```
ASW1(config)#interface po1
ASW1(config-if)#shutdown
ASW1(config-if)#do show etherchannel summary
Flags: D - down          P - bundled in port-channel
       I - stand-alone  s - suspended
       H - Hot-standby (LACP only)
       R - Layer3       S - Layer2
       U - in use       N - not in use, no aggregation
       f - failed to allocate aggregator

       M - not in use, minimum links not met
       m - not in use, port not aggregated due to minimum links not met
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

       A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SD)      LACP        Gi0/0(D)  Gi0/1(D)  Gi0/2(D)
                   Gi0/3(D)
```

Explanation: Flag D means down, shown on both the port-channel and member interfaces.

Suspended Interface Example:

```
ASW1(config)#interface g0/0
ASW1(config-if)#switchport mode access
ASW1(config-if)#do show etherchannel summary
Flags: D - down          P - bundled in port-channel
       I - stand-alone  s - suspended
       H - Hot-standby (LACP only)
       R - Layer3       S - Layer2
       U - in use       N - not in use, no aggregation
       f - failed to allocate aggregator

       M - not in use, minimum links not met
       m - not in use, port not aggregated due to minimum links not met
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

       A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)      LACP        Gi0/0(s)  Gi0/1(P)  Gi0/2(P)
                   Gi0/3(P)
```

Explanation: Lower-case s means suspended, Only G0/0 is suspended, the EtherChannel continues operating with the remaining interfaces.

SHOW ETHERCHANNEL PORT-CHANNEL

CLI Image: ASW1# show etherchannel port-channel

```
ASW1#show etherchannel port-channel
Channel-group listing:
-----
Group: 1
-----
Port-channels in the group:
-----
Port-channel: Po1 (Primary Aggregator)
-----
Age of the Port-channel = 0d:00h:36m:48s
Logical slot/port = 16/0 Number of ports = 4
HotStandBy port = null
Port state = Port-channel Ag-Inuse
Protocol = LACP
Port security = Disabled

Ports in the Port-channel:

Index Load Port EC state No of bits
-----+-----+-----+-----+-----
0 00 Gi0/0 Active 0
0 00 Gi0/1 Active 0
0 00 Gi0/2 Active 0
0 00 Gi0/3 Active 0

Time since last port bundled: 0d:00h:00m:02s Gi0/0
Time since last port Un-bundled: 0d:00h:00m:42s Gi0/0
```

Explanation:

This command shows:

- Number of ports in the port-channel
- Protocol being used
- Channel-group mode (active in this case)

Most Common Verification Command

For EtherChannel, the most commonly used command is:

#SHOW ETHERCHANNEL SUMMARY

Spanning-Tree After EtherChannel Configuration

CLI Image: ASW1# show spanning-tree

```
ASW1#show spanning-tree

VLAN0001
  Spanning tree enabled protocol rstp
  Root ID    Priority    32769
             Address     0c04.cf10.ea00
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0c04.cf10.ea00
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Po1                      Desg FWD 3         128.65 Shr
```

Explanation: Only the port-channel interface appears.
The physical interfaces do not appear at all.

Effect on Spanning-Tree:

Spanning-tree treats the four physical interfaces as one logical interface.

Instead of blocking three interfaces:

- All interfaces can forward and receive traffic
- No Layer 2 loop is formed

Key Takeaways (From This Video Section)

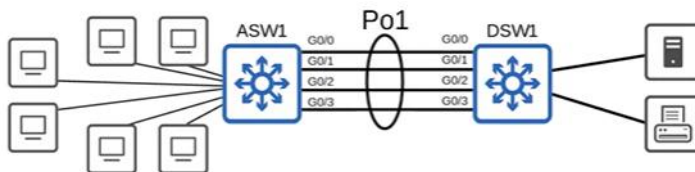
- CHANNEL-PROTOCOL exists but is rarely needed
- Protocol mismatch causes channel-group commands to fail
- Port-channel interface is configured, not individual links
- Member interfaces must match configuration
- SHOW ETHERCHANNEL SUMMARY is the most useful command
- Spanning-tree sees EtherChannel as one interface

Configuring Layer 3 (switch) EtherChannel

Layer 3 EtherChannel Introduction

let's take a brief look at Layer 3 EtherChannels.

ASW1 and DSW1 are replaced with multilayer switches.



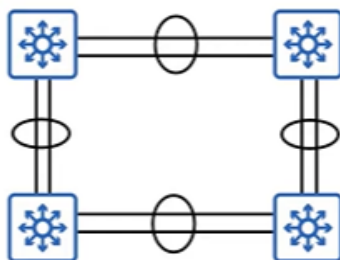
Instead of a Layer 2 connection, a Layer 3 connection is used between them.

Why Use Layer 3 Connections Between Switches

Modern network design often uses Layer 3 connections between switches, because:

- Spanning-tree won't be an issue anywhere in the network
- All interfaces can be up and forwarding
- No interfaces need to be disabled by spanning-tree

Mesh Example with Layer 3 Links



Explanation: If switches are connected using Layer 3 routed ports, all interfaces forward and none are blocked.

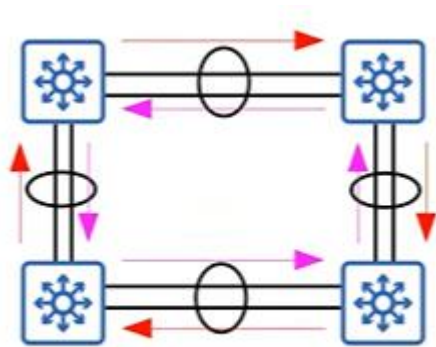
EtherChannel Does Not Fully Remove STP in Layer 2

You may think EtherChannel removes the need for spanning-tree.

However:

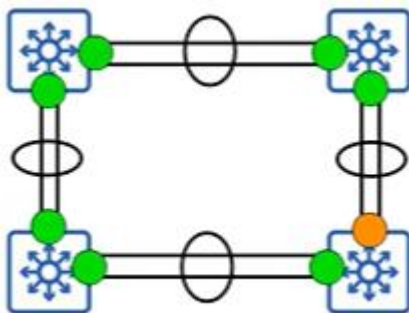
- EtherChannel only treats multiple links between two switches as one
- Layer 2 loops can still occur if multiple switches form a loop

Layer 2 Loop With EtherChannel



Explanation: Even though all links are EtherChannels, broadcasts can still loop and cause a broadcast storm.

So, Spanning-tree will block **one port-channel interface**.

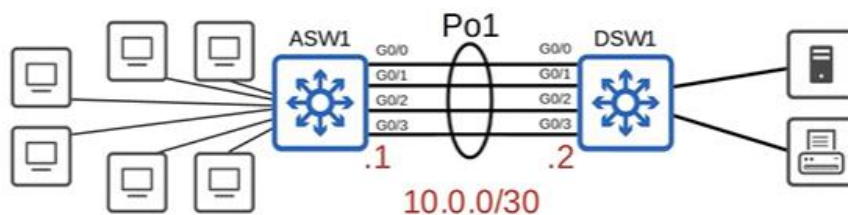


Why Layer 3 Solves This Completely

If all inter-switch links are **routed ports**:

- No Layer 2 broadcasts are forwarded
- No Layer 2 loops can be formed
- No need to run spanning-tree

Configuring a Layer 3 EtherChannel



Clean Starting Point: No port-channels are configured yet on ASW1.

Step 1: Interface Range + Routed Ports

CLI Image:

```
ASW1(config)#int range g0/0 - 3
ASW1(config-if-range)#no switchport
ASW1(config-if-range)#channel-group 1 mode active
Creating a port-channel interface Port-channel 1
```

Explanation: Member interfaces are converted to Layer 3 routed interfaces before forming the EtherChannel.

Port-Channel Auto Configuration:

If you run the command show running-config, You Notice that the port-channel interface that was created has the NO SWITCHPORT command apply on it automatically

```

interface Port-channel1
no switchport
no ip address
!
interface GigabitEthernet0/0
no switchport
no ip address
negotiation auto
channel-group 1 mode active
!
interface GigabitEthernet0/1
no switchport
no ip address
negotiation auto
channel-group 1 mode active
!
interface GigabitEthernet0/2
no switchport
no ip address
negotiation auto
channel-group 1 mode active
!
interface GigabitEthernet0/3
no switchport
no ip address
negotiation auto
channel-group 1 mode active
!

```

Explanation: The port-channel interface automatically receives the no switchport command.

IP Address for Layer 3 EtherChannel:

Since this is a Layer 3 EtherChannel, an IP address is required.

Where to Configure It

It is configured on the port-channel interface.

CLI Image:

```

ASW1(config-if-range)#int po1
ASW1(config-if)#ip address 10.0.0.1 255.255.255.252
ASW1(config-if)#

```

Explanation: The IP address is assigned to the port-channel interface.

Layer 3 EtherChannel Topology:

ASW1 (10.0.0.1) === Po1 === (10.0.0.2) DSW1

10.0.0.0/30

Explanation: ASW1 and DSW1 are now connected like two routers at Layer 3.

Verifying Layer 3 EtherChannel:

CLI Image:

```
ASW1(config-if)#do sh etherch sum
Flags:  D - down          P - bundled in port-channel
        I - stand-alone  S - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG
```

```
Number of channel-groups in use: 1
Number of aggregators:          1
```

Group	Port-channel	Protocol	Ports
1	Port (RU)	LACP	Gi0/0(P) Gi0/1(P) Gi0/2(P) Gi0/3(P)

Explanation:

- R = Layer 3 EtherChannel (routed port)
- U = In use

This replaces the S flag used for Layer 2 EtherChannels.

IP Interface Verification:

CLI Image:

```
ASW1#show ip interface brief
Interface          IP-Address      OK? Method Status  Protoc
GigabitEthernet0/0 unassigned      YES manual up       up
GigabitEthernet0/1 unassigned      YES manual up       up
GigabitEthernet0/2 unassigned      YES manual up       up
GigabitEthernet0/3 unassigned      YES manual up       up
GigabitEthernet1/0 unassigned      YES unset  up       up
GigabitEthernet1/1 unassigned      YES unset  up       up
GigabitEthernet1/2 unassigned      YES unset  up       up
GigabitEthernet1/3 unassigned      YES unset  up       up
GigabitEthernet2/0 unassigned      YES unset  up       up
GigabitEthernet2/1 unassigned      YES unset  up       up
GigabitEthernet2/2 unassigned      YES unset  up       up
GigabitEthernet2/3 unassigned      YES unset  up       up
GigabitEthernet3/0 unassigned      YES unset  up       up
GigabitEthernet3/1 unassigned      YES unset  up       up
GigabitEthernet3/2 unassigned      YES unset  up       up
GigabitEthernet3/3 unassigned      YES unset  up       up
Port-channel1      10.0.0.1        YES NVRAM  up       up
ASW1#
```

Explanation: The IP address appears only on the port-channel interface.

Effect on Spanning-Tree:

- Spanning-tree is not running on the connection
- ASW1 and DSW1 communicate at Layer 3
- No Layer 2 broadcasts
- No Layer 2 loops

Load Balancing Still Works:

Just like Layer 2 EtherChannel

- Traffic is load-balanced across all four member interfaces

Final Takeaway

- Layer 3 EtherChannel removes spanning-tree from inter-switch links
- Port-channel interface acts like a routed interface
- Load balancing still applies
- Understanding purpose, setup, and verification is enough for CCNA

Final Command Review

Configure Load-Balancing

SW(config) `port-channel load-balance mode`

#configures the EtherChannel load-balancing method on the switch

View Load-Balancing

SW# `show etherchannel load-balance`

#displays information about the load-balancing settings

Create EtherChannel

SW(config-if)# `channel-group number mode {desirable|auto|active|passive|on}`

#configures an interface to be part of an EtherChannel

Most Useful Verification Command

SW# `show etherchannel summary`

#displays a summary of EtherChannels on the switch

Detailed Port-Channel Information

SW# `show etherchannel port-channel`

#displays information about the virtual port-channel interfaces on the switch

The following table displays the channel-group configurations that will establish an EtherChannel:

SwitchB	off	auto	desirable	passive	active	on
SwitchA						
off	NO	NO	NO	NO	NO	NO
auto	NO	NO	PAgP	NO	NO	NO
desirable	NO	PAgP	PAgP	NO	NO	NO
passive	NO	NO	NO	NO	LACP	NO
active	NO	NO	NO	LACP	LACP	NO
on	NO	NO	NO	NO	NO	ON